An Intro to Big Data and Machine Learning for Survey Researchers and Social Scientists

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SVY 751, Notes Part 0

Course Objectives

- This course will offer participants:
  - an overview of key Big Data terminology and concepts
  - an introduction to common data generating processes
  - a discussion of some primary issues with linking Big Data with Survey Data
  - issues of coverage and measurement errors within the Big Data context
  - a discussion of information extraction and signal detection in the context of Big Data
  - a discussion of the similarities and differences in model building for inference versus prediction
  - an overview of general concepts from machine learning as they apply to processing Big Data
  - a discussion of signal detection and information extraction
  - a discussion of the potential pitfalls for inference from Big Data
  - an introduction to a small set of key analytic techniques (e.g. classification trees, random forests, conditional forests) to process Big Data using R with example code provided

Outline for the Course

- Overview of Big Data
  - What is Big Data?
  - Why does it matter?
  - What's in this for Survey Researchers?
- Working with Big Data
  - Compilation
    - **Sources/Data Bases**
  - Computing Structures/Infrastructure
    - **Cloud Computing, Parallel Processing**
  - Cautions and Limitations
    - **Total Survey Error for Big Data**
    - **Sources of error (e.g. merging, compiling, measurement)**
    - **Analytic limitations – projectability…Does Big=Better?!**

Outline for the Course, Cont.

- Analyses of Big Data
  - Classical Statistical Approaches versus Statistical Machine Learning
  - Model Evaluation/Validation
  - Popular machine learning techniques
    - **Clustering**
      - K-means
      - Hierarchical
    - **K Nearest Neighbors**
    - **CARTS**
    - **Random Forests**

Overview of Big Data

What is Big Data?

Why does it matter?

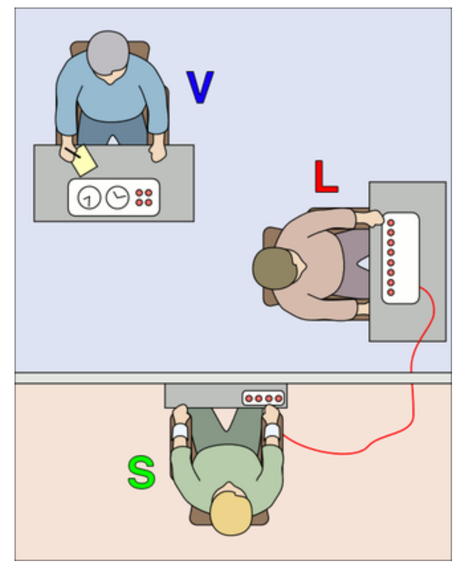
What’s in this for Survey Researchers?

Until Recently: Three main sources

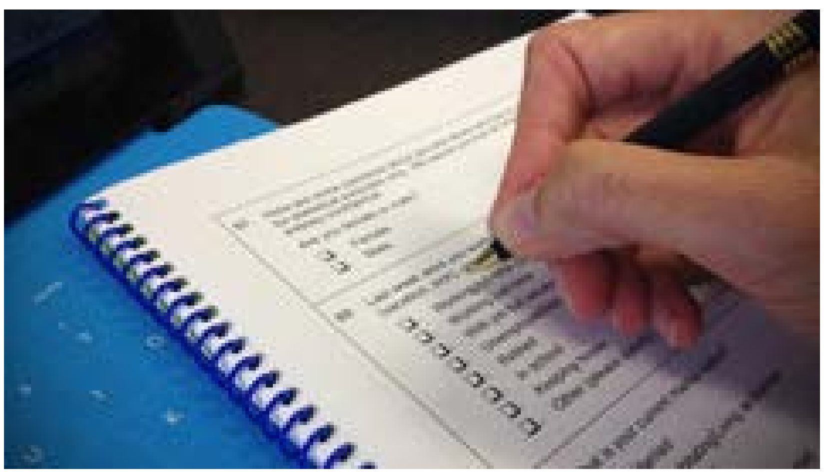
Administrative Data


2
3 Steuernummer
eTIN lt. Lohnsteuerbescheinigung(en), sofern vorhanden
eTIN lt. weiterer Lohnsteuerbeschei
4
Einkünfte aus nichtselbständiger Arbeit
Angaben zum Arbeitslohn
Lohnsteuerbescheinigung(en) Steuerklasse 1 - 5
5 Steuerklasse 168
EUR Ct
6 Bruttoarbeitslohn 110
7 Lohnsteuer 140
8 Solidaritätszuschlag 150

Experiments



Survey Data



Now

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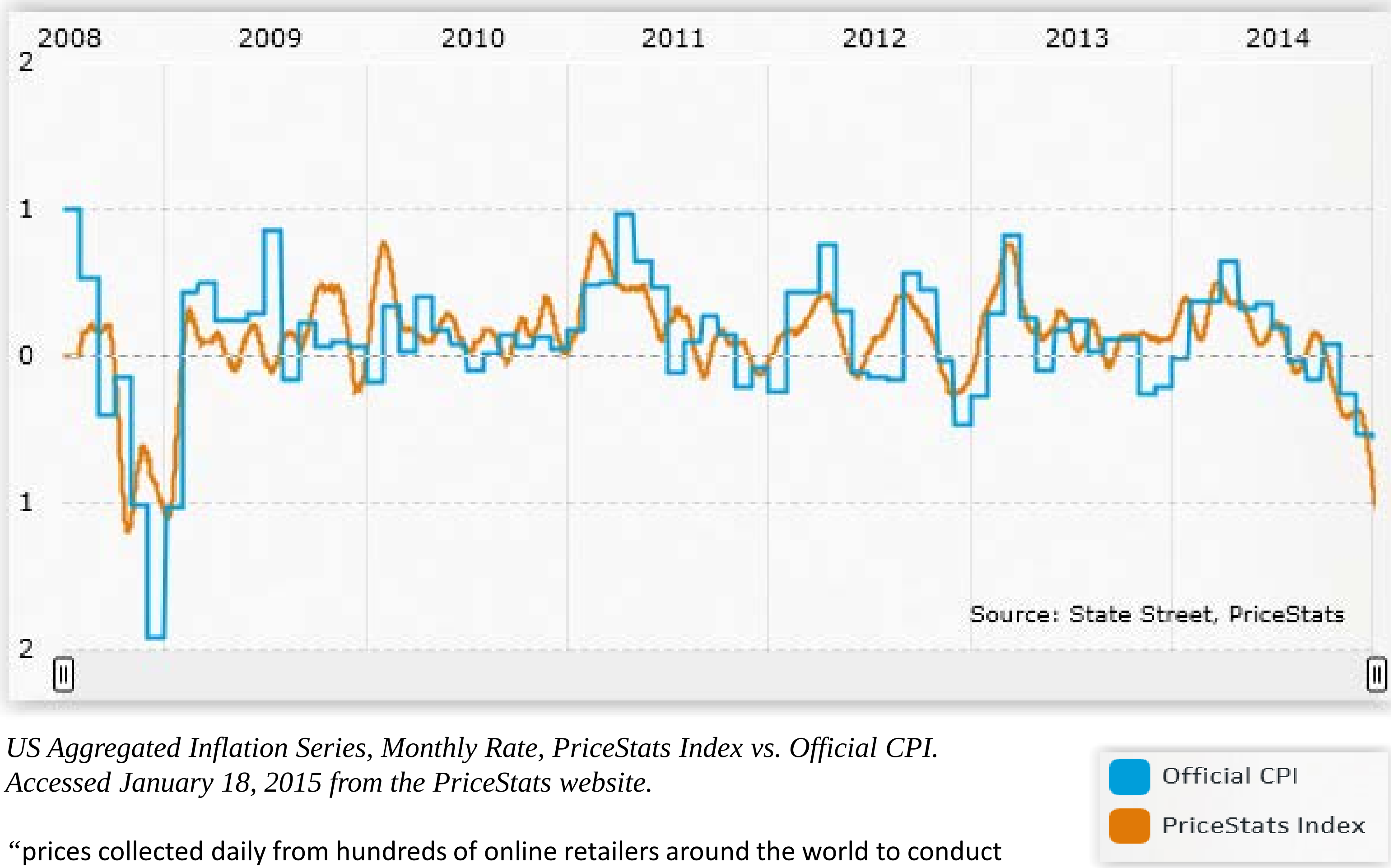
MIT Billion Prices Projects, PriceStats



US Aggregated Inflation Series, Monthly Rate, PriceStats Index vs. Official CPI. Accessed January 18, 2015 from the PriceStats website.

"prices collected daily from hundreds of online retailers around the world to conduct economic research"

Google Flue Trend http://dx.doi.org/10.1038/nature07634

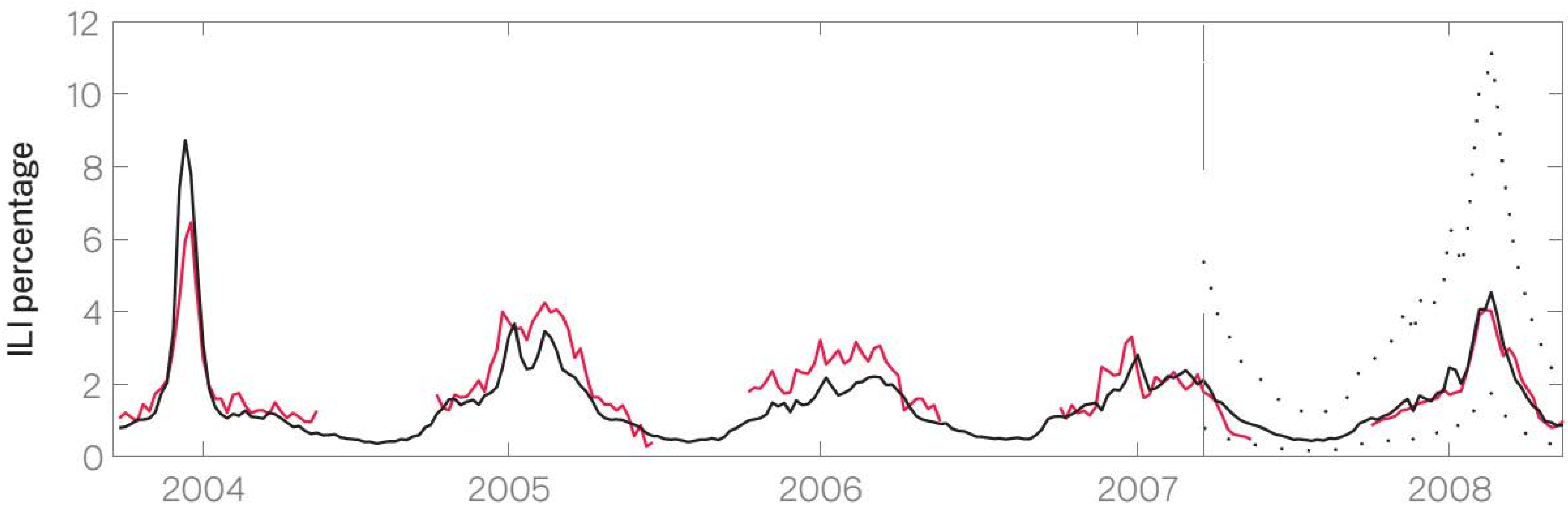


Figure 2: A comparison of model estimates for the Mid-Atlantic Region (black) against CDC-reported ILI percentages (red), including points over which the model was fit and validated. A correlation of 0.85 was obtained over 128 points from this region to which the model was fit, while a correlation of 0.96 was obtained over 42 validation points. 95% prediction intervals are indicated.

Notes

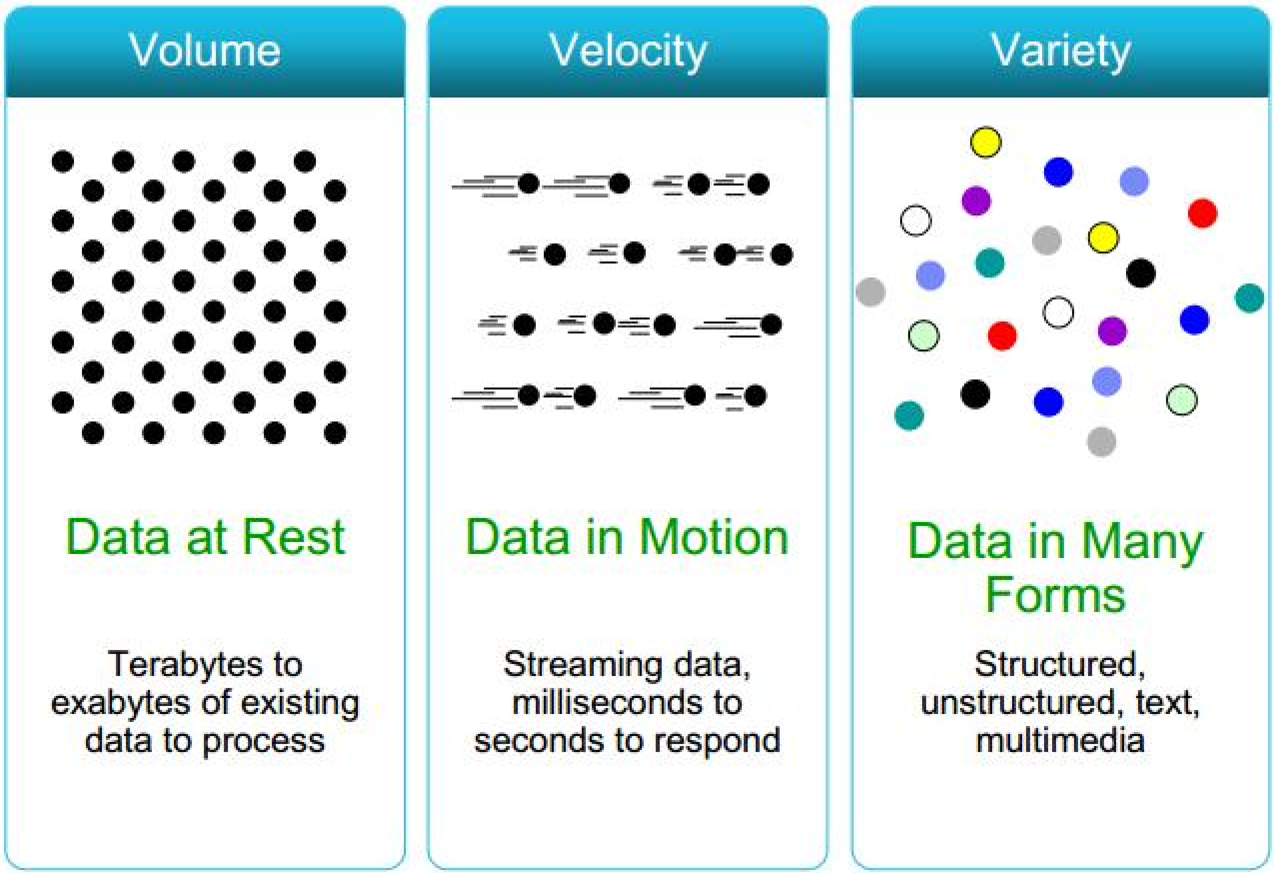
Types of Big Data

There are many different types of Big Data sources:

- Social media data
- Personal data (e.g. data from tracking devices)
- Sensor data
- Transactional data
- Administrative data

They share features …

Big Data - Features



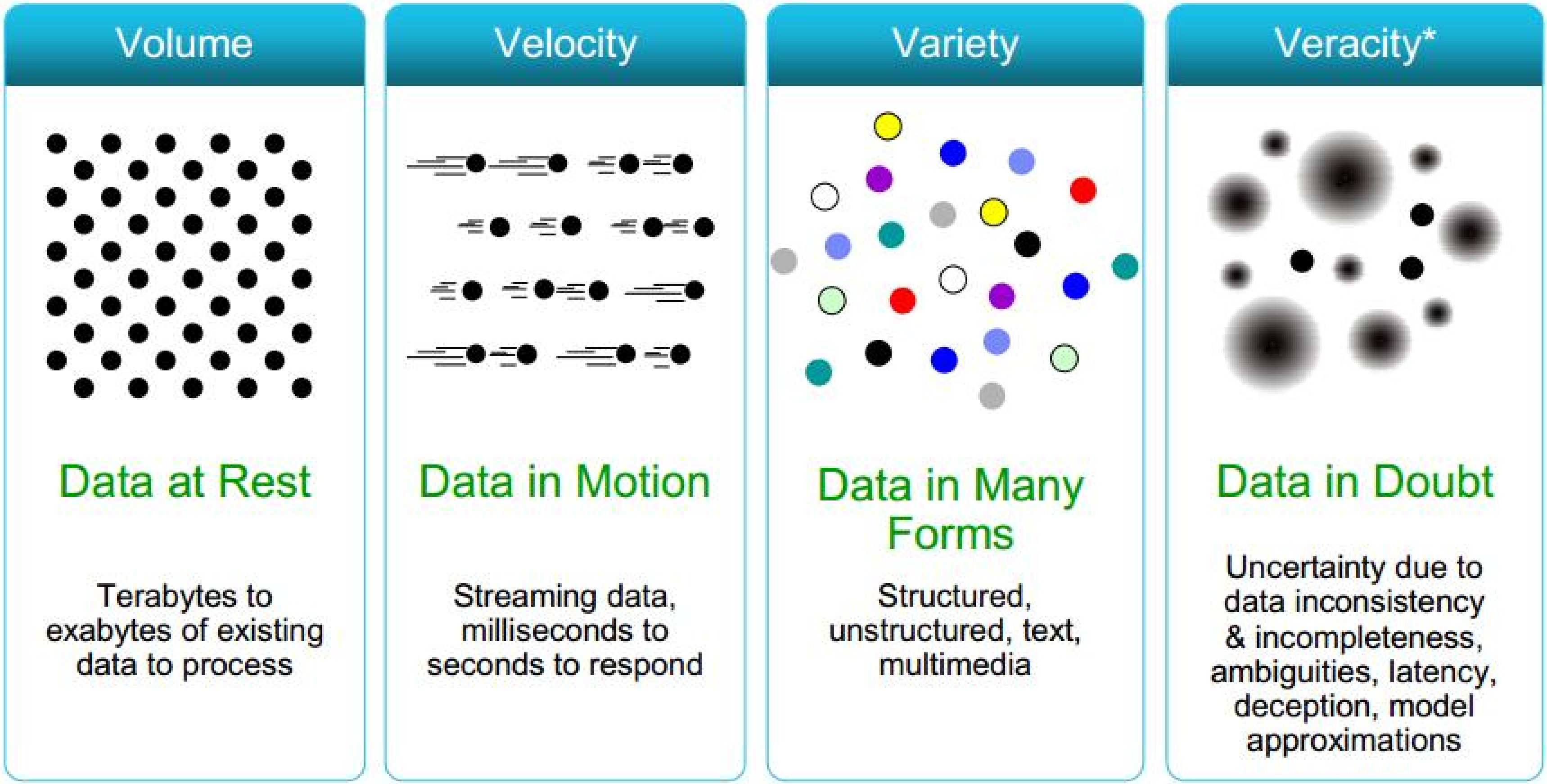
Hope

Expensive data collections can be replaced

More (= better) data for decision making

Information available in (nearly) real time

But (at least) one more V



http://www.rosebt.com/blog/data-veracity

Veracity

Who? What? Why?

Who is missing? Who is counted repeatedly?

What is not said / measured? ..and why?

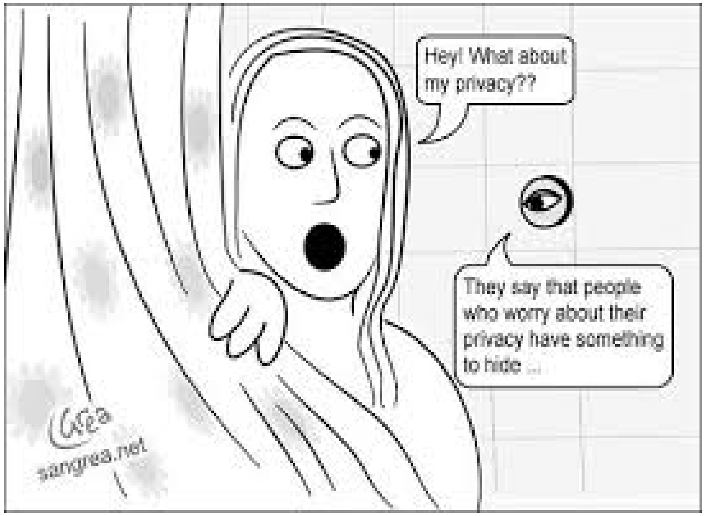
Tensions

- Privacy vs. Public Good
- Convenience vs. Accusation

"Your recent Amazon purchases, Tweet score and location history makes you 23.5% welcome here."

Tensions

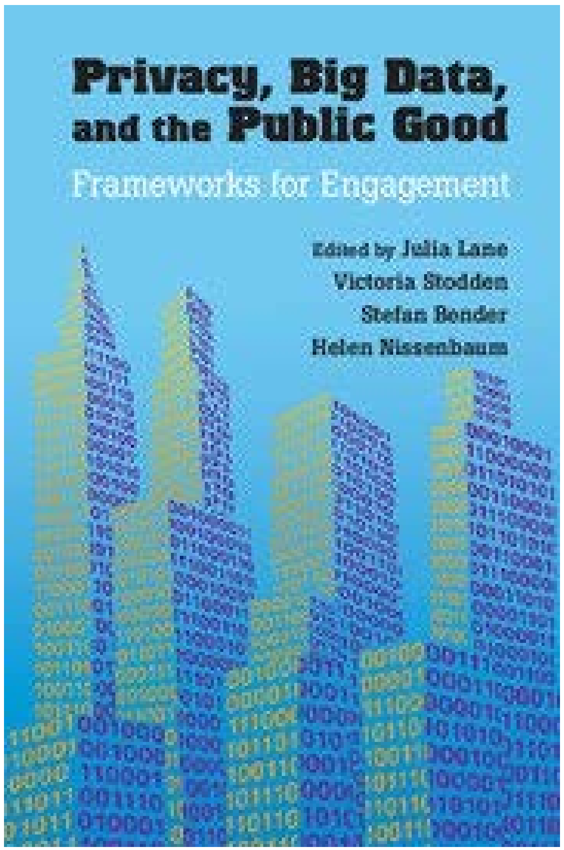
- Privacy vs. Public Good
- Convenience vs. Accusation
- Identifiable vs. Reachable
- "Tyranny of Minority"



Tensions

- Privacy vs. Public Good
- Convenience vs. Accusation
- Identifiable vs. Reachable
- "Tyranny of Minority"
- Consent vs. Confusion
- Privacy vs. Data quality
- Europe vs. USA
- "Big Brother" vs. "Big Mother"

Good Discussion



- What are the legal requirements?
- What are the rules of engagement?
- What are the best ways to provide access while also protecting confidentiality?
- Are there reasonable mechanisms to compensate citizens for privacy loss?
- How can we built trustworthy curators?
- What do we (need to) know about the data generating process?
- How can we increase linkage without increasing risks?

‘Big Data’ – you already know

- \- Paradata (ideally)
  - real time, longitudinal, “everybody”
- \- Useful for survey researchers to monitor
  - inactive interviewer
  - coverage composition
- \- Allows action while it happens to save cost
- \- Even more opportunity in web surveys
  - mouse movements
  - response times
  - missing data

Working with Big Data

Compilation: Sources/Data Bases

Cautions and Limitations: Total Error for Big Data

- Sources of error (e.g. merging, compiling, measurement)
- Analytic limitations – projectability…Does Big=Better?! (Reach versus Relevance)

Computing Structures/Infrastructure: Cloud Computing, Parallel Processing

Key Ingredients for Valid Inference

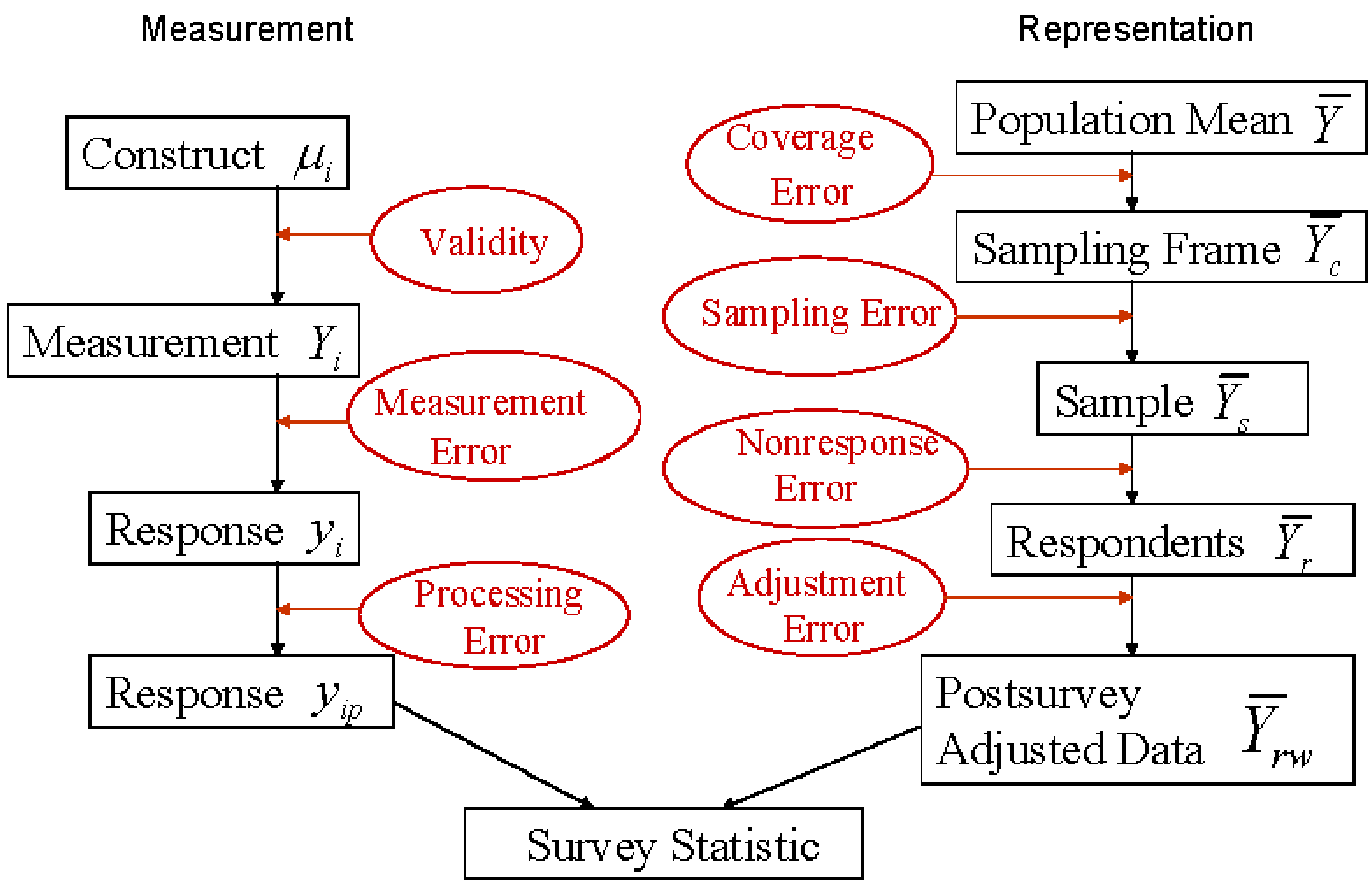
1. Data generating process needs to be known
2. Error framework already available for small data .. expand!
3. Data access and linkage is essential part

1. Data Generating Process

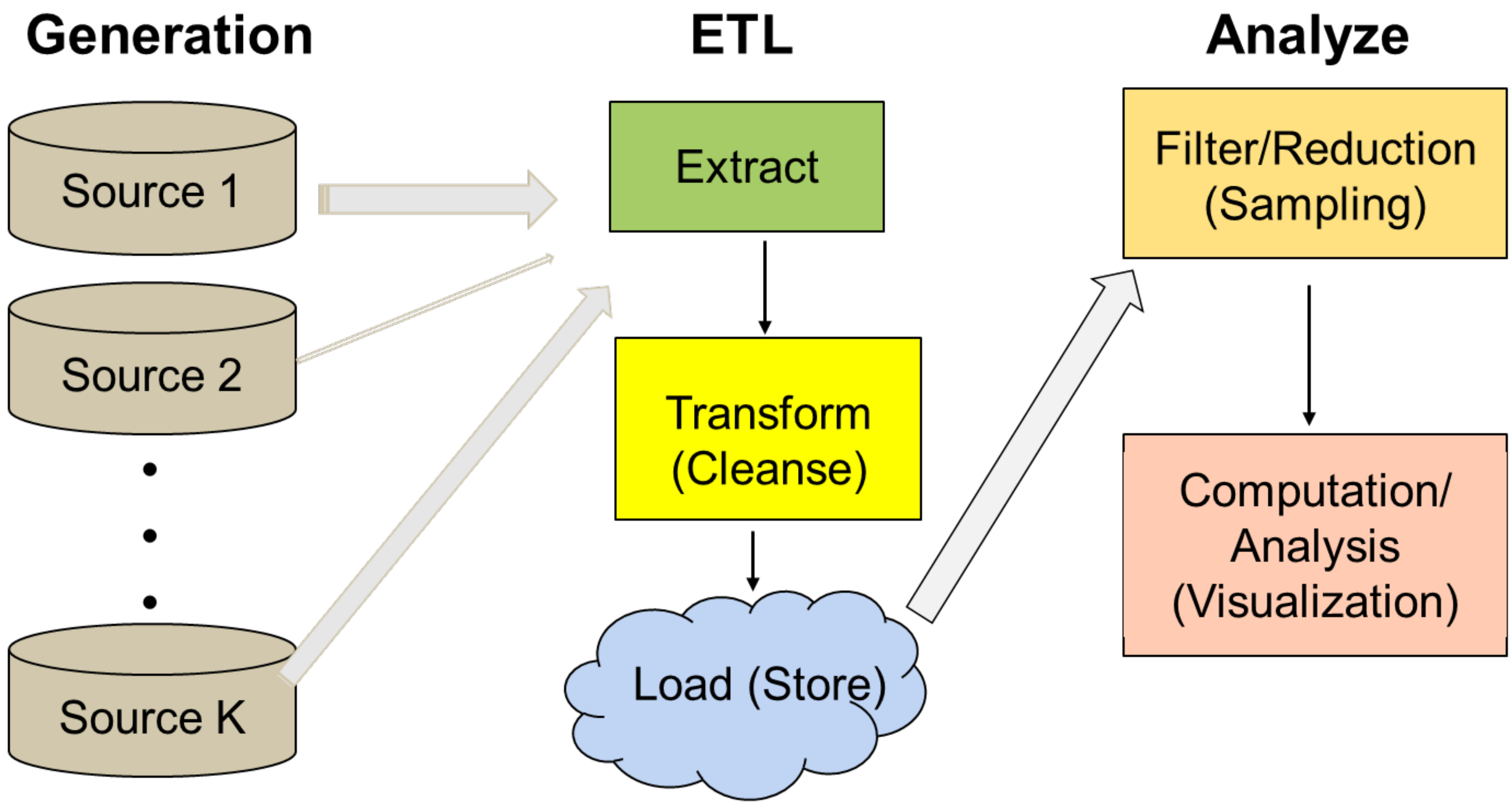
What makes working with Big Data different is the found nature of the data

Organic vs. Design

2. Error Framework for Surveys



2. cnt'd Extended Error Model (Biemer 2014)



3. Access and Linkage

Essential to understand data quality and break-downs

Challenged by … different privacy requirements

- Open issues of ownership
- Lack of trusted third parties
- Ill-defined legal space
  consent: either comprehensive or comprehensible

However …. likely leads to good data documentation

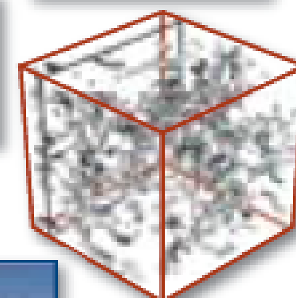
+ Reproducible research
+ Transparency

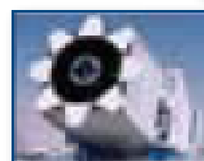
Paradigm Shift

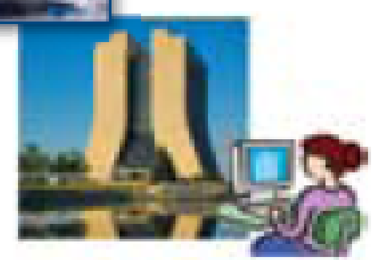
Science Paradigms

- Thousand years ago:
  science was **empirical**
  describing natural phenomena
- Last few hundred years:
  theoretical branch
  using models, generalizations
- Last few decades:
  a **computational** branch
  simulating complex phenomena
- Today: **data exploration** (eScience)
  unify theory, experiment, and simulation
  – Data captured by instruments or generated by simulator
  – Processed by software
  – Information/knowledge stored in computer
  – Scientist analyzes database/files using data management and statistics

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{4\pi G p}{3} - K\frac{c^2}{a^2}$$







Hey et al. (2009).

Skills

RESEARCHER
Team member with experience applying formal research methods, including survey methodology and statistics
DOMAIN EXPERT
User, analyst, or leaders with deep subject matter expertise related to the data, its appropriate use, and its limitations
COMPUTER SCIENTIST
SYS ADMIN
Team member responsible for defining and maintaining a computation infrastructure that enalbes large scale computation

Computing Environment

- Apache Hadoop. A system for maintaining a distributed file system that supports the storage of large-scale (Terabytes or Petabytes of content), and the parallel processing of algorithms against large data collections. Requires a programming language such as Java or Python.
- Apache Spark. A fast and general purpose engine for large-scale data processing that works in support of Hadoop or in-memory databases. Requires a programming language such as Java or Python.
- Java programming language. A general purpose systems engineering language that supports the creation of efficient algorithms for data analysis.
- Python programming language. A general purpose systems engineering language that supports rapid prototyping and efficient algorithms for data analysis.

Example: *Amazon Elastic MapReduce (EMR)*